

Ji-Zhe Zhang

Postdoctoral Researcher · Institute of Chemistry, Chinese Academy of Sciences

jizhe0726@iccas.ac.cn

ORCID: 0009-0001-7825-9042

github.com/zhangjizhe2013-netizen

Research Profile

Experimental researcher working across integrated photonics, precision weak-light detection, and molecular spin quantum sensing. My doctoral research established a foundation in thin-film lithium-niobate photonic devices, high-frequency acousto-optic and electro-optic control, and reconfigurable photonic structures. My current postdoctoral work investigates excited-state spins in organic molecular crystals, spin-dependent optical readout, magneto-optical effects, and weak-field sensing.

Education & Appointments

Institute of Chemistry, Chinese Academy of Sciences

2025 — Present

Postdoctoral Researcher · Chuang Zhang Group

- Excited-state spin properties of organic molecular crystals and their optical readout.
- Host-guest interactions for enhanced triplet-spin optical signals and precision weak-field sensing.
- Weak-fluorescence detection, magneto-optical characterization, and microwave-enabled optical measurements.

University of Science and Technology of China

Degree awarded 2025

Ph.D. in Physics

- Thin-film lithium-niobate integrated photonic chips and on-chip optical-field control.
- High-frequency acousto-optic and electro-optic modulation, optical delay, and reconfigurable photonic structures.
- Micro/nanophotonic device design and fabrication, optical setup construction, RF/microwave control, and precision characterization.

Selected Publications

1. **Ji-Zhe Zhang**[†], Xin-Biao Xu[†], Yanjun Gong[†], et al. "Optically-driven organic nano-step actuator for reconfigurable photonic circuits." *Nature Communications* **16**, 8213 (2025). doi:10.1038/s41467-025-63521-z. **Co-first author**
2. Zhengwei Yang, Yu Zhang, Qingda Chang, Shuchun Zhang, **Ji-Zhe Zhang**^{*}, Chuang Zhang^{*}, Jiannian Yao. "Magnetic Field Induced Circular Polarization of Photoluminescence from Manganese Doped Perovskites." *Advanced Optical Materials* **13**(27), e00560 (2025). doi:10.1002/adom.202500560. **Co-corresponding author**
3. **Ji-Zhe Zhang**, Yu Zeng, Qing Qin, Yuan-Hao Yang, Zheng-Hui Tian, et al. "On-chip 7 GHz acousto-optic modulators for visible wavelengths." *Optics Express* **33**(3), 5562-5576 (2025). doi:10.1364/OE.540356. **First author**
4. **Ji-Zhe Zhang**, Jia-Lin Chen, Jia-Wen Fan, Jia-Qi Wang, Yuan-Hao Yang, et al. "Electro-optically tunable optical delay on a lithium niobate photonic chip." *Optics Letters* **49**(13), 3737-3740 (2024). doi:10.1364/OL.519479. **First author**

[†] Equal contribution; ^{*} corresponding author.

Research Highlights

7 GHz Visible-frequency on-chip acousto-optics Demonstrated high-frequency acousto-optic phase modulation on a lithium-niobate-on-sapphire platform for visible wavelengths.	30 nm Programmable nano-step actuation Developed optical control and precision-displacement measurement for an organic actuator used to reconfigure photonic structures.	Spin-light Weak-signal optical readout Investigating host-mediated preparation and optical readout of molecular triplet spins for precision magnetic sensing.
--	--	---

Selected Research Experience

- 01

Molecular excited-state spin optical readout and weak-field sensing
Study spin-dependent photophysics in organic molecular crystals, using host-guest interactions to enhance optical readout of triplet states. Develop weak-fluorescence and magnetic-field-response measurements for precision sensing and future integration with micro/nanophotonic structures.
- 02

Thin-film lithium-niobate photonic devices
Designed, fabricated, and characterized integrated photonic devices for 7 GHz visible acousto-optic modulation, electro-optically tunable coupling, and on-chip optical delay.
- 03

Optically driven micro-actuators for reconfigurable photonics
Built an optical-actuation experiment and precision-displacement measurement platform, achieving programmable approximately 30 nm steps and post-fabrication tuning of on-chip photonic structures.

Research Skills

Integrated photonics	Thin-film lithium-niobate devices, waveguides and microcavities, acousto-optic/electro-optic control, on-chip optical testing
Micro/nano experiments	Device design and fabrication, packaging and coupling, micro-scale optical manipulation, precision displacement measurement
Precision measurement	Weak-fluorescence detection, low-noise optics, magneto-optical characterization, weak-field sensing
Spin & spectroscopy	Molecular excited-state spin optical readout, magnetic-field-dependent spectroscopy, polarization measurements
Experimental systems	Custom optical setups, RF and microwave control, data acquisition, and system integration

Future Research Directions

Integrated enhancement of weak optical signals Waveguides, microcavities, and high-speed modulation for efficient collection and low-noise readout.	On-chip readout of molecular spins Coupling organic spin systems to integrated photonic structures for compact, high-contrast magnetic sensing.	Chip-scale precision optoelectronic measurement Integrating excitation, control, sensing, synchronous detection, and feedback in a unified platform.
---	---	--